

Algebra II with Trigonometry

Chapter 5.5

Study guide

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Certainly! Here is a comprehensive and self-contained study guide and summary based on your document.

Law of Sines Study Guide

1. Short Summary of the Document

This document is an instructional lesson on the **Law of Sines**, a fundamental trigonometric rule for solving oblique (non-right) triangles. It covers when and how to use the Law of Sines, especially in cases where triangles are not right-angled. The guide details the four types of triangle cases, placing particular emphasis on ambiguous SSA (Side-Side-Angle) triangles, and provides strategies for determining the number of possible triangle solutions (none, one, or two). It includes several example problems and word problems applying the Law of Sines.

2. Core Definitions, Theorems, and Formulas

Triangle Cases:

- **Case 1 (ASA/SAA):** One side and two angles are known.
- **Case 2 (SSA):** Two sides and the angle opposite one of those sides are known (**ambiguous case**).
- **Case 3 (SAS):** Two sides and the included angle are known.
- **Case 4 (SSS):** All three sides are known.

The Law of Sines

For a triangle ABC with sides a, b, c opposite angles A, B, C respectively:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Or rearranged:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

When to Use:

Use the Law of Sines to solve triangles in Cases 1 (ASA/SAA) and 2 (SSA).

The Ambiguous (SSA) Case

Depending on the information given (for SSA), there can be:

- **No solution:** No triangle fits the data.
- **One solution:** Exactly one triangle fits.
- **Two solutions:** Two different triangles fit.

Calculating the "height":

$$h = b \sin A$$

For $A < 90^\circ$ (Acute Angle) 1. **If $b > a > b \sin A$:** Two solutions 2. **If $a > b > b \sin A$:** One solution 3. **If $a < b \sin A < b$:** No solution

For $A \geq 90^\circ$ (Obtuse Angle) 1. **If $a > b$:** One solution 2. **If $a \leq b$:** No solution

3. Step-by-Step Study Guide

Step 1: Identify the Known Information

- Check if you have SSA, ASA, SAA, SAS, or SSS.
- For **ASA/SAA:** Use Law of Sines directly.
- For **SSA:** Check for ambiguous cases.

Step 2: Write the Law of Sines

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Step 3: Set Up the Equation

- If solving for a side:

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

- If solving for an angle:

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

then solve for $\sin B$ or $\sin A$.

Step 4: For SSA, Check for the Number of Solutions

1. Calculate the height $h = b \sin A$. 2. Compare given sides a , b and h :

- **If $a < h$, no triangle.**
- **If $a = h$, one right triangle.**
- **If $h < a < b$, two triangles.**
- **If $a \geq b$, one triangle.**
- For obtuse angle, only one triangle if $a > b$, none if $a \leq b$.

Step 5: Solve for Missing Parts

- Use the Law of Sines to solve for unknown angles or sides.
- Remember, $\sin^{-1}$ might give two possible angle values.
- Check if the sum of the triangle's angles is less than 180° .

Step 6: Answer Word Problems

- Draw a diagram.
- Label all known sides and angles.
- Identify which method is appropriate.
- Proceed as above.

4. Practice Problems With Answers

1. In $\triangle ABC$, $A = 40^\circ$, $B = 75^\circ$, $a = 30$. Find side b .

Answer: First, find $C = 180^\circ - 40^\circ - 75^\circ = 65^\circ$.

$$\begin{aligned}\frac{a}{\sin A} &= \frac{b}{\sin B} \\ \frac{30}{\sin 40^\circ} &= \frac{b}{\sin 75^\circ} \\ b &= \frac{30 \cdot \sin 75^\circ}{\sin 40^\circ} \approx \frac{30 \cdot 0.9659}{0.6428} \approx 45.1\end{aligned}$$

2. Solve triangle with $a = 12$, $b = 20$, $A = 25^\circ$. How many solutions? Find B for the possible case(s).

Answer: Height: $h = b \sin A = 20 \times \sin 25^\circ \approx 8.45$ Since $a = 12 > h$ but $a < b$, **two triangles possible.**

$$\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = b \frac{\sin A}{a} = 20 \frac{\sin 25^\circ}{12} \approx 0.7058$$

$$B_1 = \sin^{-1}(0.7058) \approx 44.9^\circ$$

$$B_2 = 180^\circ - 44.9^\circ = 135.1^\circ$$

Check the sum for each with A :

- $A + B_1 = 69.9^\circ < 180^\circ$, triangle possible.
- $A + B_2 = 160.1^\circ < 180^\circ$, triangle possible.

Both solutions are valid.

3. Given $b = 15$, $c = 25$, $B = 25^\circ$. Find all possible values for angle C .

Answer:

$$\frac{\sin C}{c} = \frac{\sin B}{b} \Rightarrow \sin C = c \frac{\sin B}{b} = 25 \frac{\sin 25^\circ}{15} \approx 0.7046$$

$$C_1 = \sin^{-1}(0.7046) \approx 44.8^\circ$$

$$C_2 = 180^\circ - 44.8^\circ = 135.2^\circ$$

Check if possible: $B + C_2 = 160.2^\circ < 180^\circ$, so two solutions. **Possible values:** $C = 44.8^\circ$ and 135.2° .

4. Solve triangle with $A = 60^\circ$, $a = 10$, $b = 11$.

Answer: $h = b \sin A = 11 \times \sin 60^\circ \approx 9.53$ Since $a = 10 > h$ and $a < b$, two triangles possible.

$$\sin B = \frac{b \sin A}{a} = \frac{11 \sin 60^\circ}{10} \approx 0.953$$

$$B_1 = \sin^{-1}(0.953) \approx 72.3^\circ \quad B_2 = 180^\circ - 72.3^\circ = 107.7^\circ$$

Both are possible.

5. A triangle has sides $a = 24$, $b = 30$, and angle $A = 90^\circ$. Find angle B .

Answer:

$$\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = b \frac{\sin A}{a} = 30 \times \frac{1}{24} = 1.25$$

But $\sin B > 1$, **no triangle exists**.

5. Assumptions & Ambiguities

- Where diagrams or some numbers were missing/unclear, the problem framework and reasoning were explicitly described based on the text.
- Some problems involve SSA ambiguous cases. The method follows the height-check and angle computation described in the document.
- Some interpretation is based on standard mathematical procedures because a few examples in the PDF are images, but all practice is kept faithful to the document's principles.
- Degrees are used throughout, assuming all angles are in degrees unless otherwise stated.

End of Study Guide